

Java Identifiers

The naming system of your code

1. What are Identifiers?

Identifiers are the unique names you give to **variables, methods, classes, and objects**. They allow you to refer to these elements throughout your program.

The Basic Syntax Rules:

- Can include: Letters (A-Z, a-z), Digits (0-9), Underscores (_), and Dollar Signs (\$).
- **Must** start with a Letter, _, or \$.
- Cannot contain spaces or special symbols (like @, #, %).
- Cannot be a Java "Reserved Word" (like `int`, `public`, or `class`).

⚠ Case Sensitivity:

Java is strictly case-sensitive. `myVar`, `myvar`, and `MYVAR` are **three different** variables.

2. Valid vs. Invalid Examples

Name	Status	Reason
<code>total_marks</code>	VALID	Uses allowed underscore.
<code>4th_place</code>	INVALID	Cannot start with a digit.

user name	INVALID	Spaces are forbidden.
\$price	VALID	\$ sign is allowed.
while	INVALID	Reserved keyword.

3. Professional Naming Conventions

While Java rules tell you what is allowed, conventions tell you what is professional.

Variables & Methods: Use **camelCase** (First word lowercase, others capitalized).

```
int studentAge; | void calculateTotal();
```

Classes: Use **PascalCase** (All words capitalized).

```
public class BankAccount { ... }
```

Constants: Use **UPPER_CASE** with underscores.

```
final int MAX_HEIGHT = 100;
```

4. Meaningful Names: Why it Matters

Code is read more often than it is written. Descriptive names tell a story.

Confusing Style:

```
int a = 60;  
int b = a * 24;
```

Pro Style:

```
int minsInHour = 60;  
int minsInDay = minsInHour *  
24;
```

Summary Checklist

- ✓ Never start with a number.
- ✓ No spaces or hyphens.
- ✓ Avoid Java keywords (abstract, switch, float, etc).
- ✓ Use descriptive words, even if they are longer.
- ✓ Stick to camelCase for variables.